

034IN - FONDAMENTI DI AUTOMATICA -
FUNDAMENTALS OF AUTOMATIC CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION
A.Y. 2023/2024

February 13, 2025

First and Last Name:

Group: Group A

Exercise: Exercise 1 - 2 Questions

Available Time Frame: 60 minutes

Important Notes:

- Write your answers on a single white sheet of paper using a black pen.
- Do not write in blue ink or pencil.
- Do not hand in additional sheets.
- The clarity and precision of your answers will be subject to evaluation.

I declare that the answers to this exercise are my own and only my own work and that I have not consulted with others.

034IN - FONDAMENTI DI AUTOMATICA -
FUNDAMENTALS OF AUTOMATIC CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION – A.Y. 2023/2024
February 13, 2025

First and Last Name:

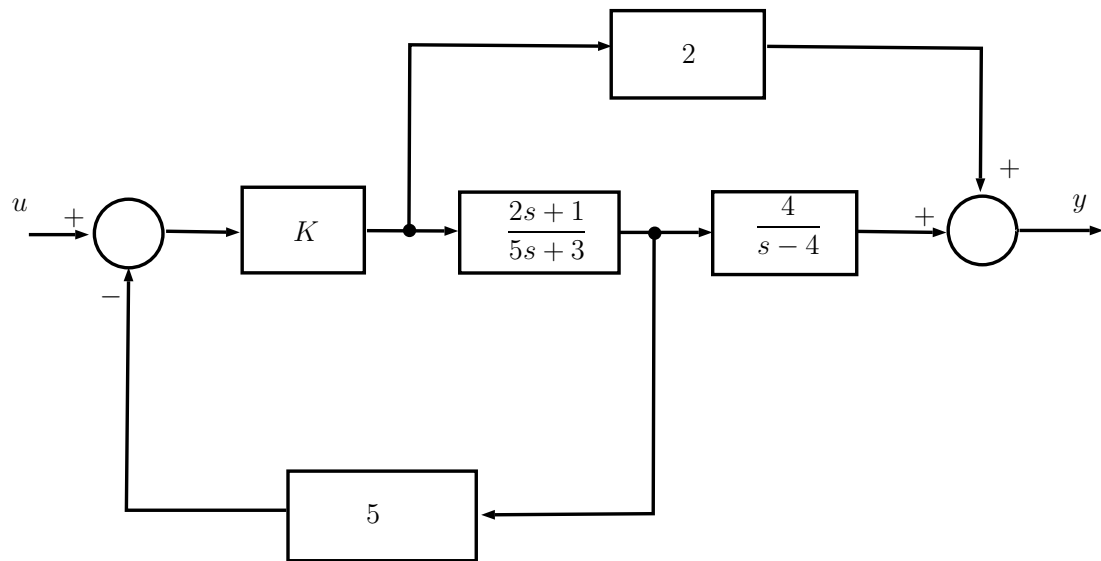
Group: Group A

Exercise: Exercise 1

Available Time Frame: 60 minutes

Exercise 1

The following block diagram describes a linear time-invariant system:



Question 1.1.

- **Determine** a representation of the LTI system using **state equations**.
- **Validate or invalidate** the following sentence, simply **by analysing the block diagram's layout**:
 - It is possible to stabilise the whole system by applying a proper value of the parameter K .

034IN - FONDAMENTI DI AUTOMATICA -
FUNDAMENTALS OF AUTOMATIC CONTROL
ADDITIONAL PARTIAL WRITTEN EXAMINATION – A.Y. 2023/2024
February 13, 2025

First and Last Name:

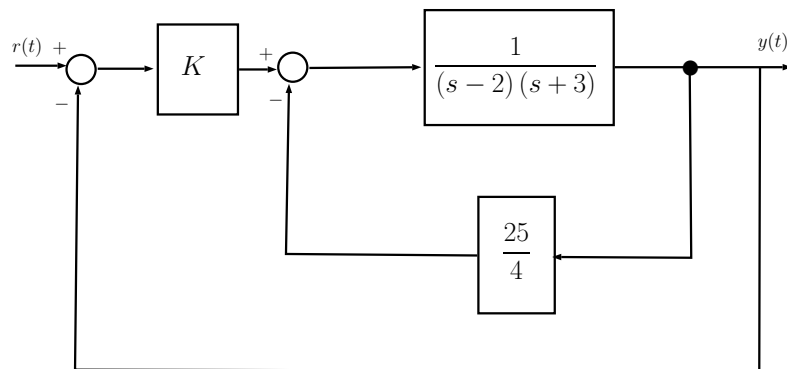
Group: Group A

Exercise: Exercise 2

Available Time Frame: 60 minutes

Exercise 2

Consider the LTI system described by the following block diagram:



where $K \in \mathbb{R}$.

Question 2.1.

- Analyse the system's stability as a function of the parameter K .
- What are the roots of the closed-loop polynomial $p(s)$, when $K = 0$?
- Explain why there is no value of K to impose that the real part of the closed-loop poles is less than $-1/2$.

Comment on your findings.